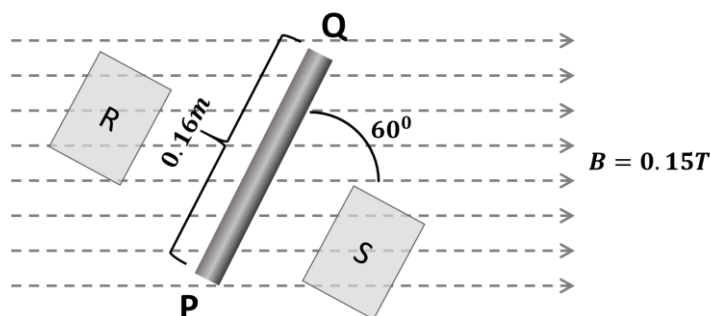


(Answer Question No. 1 and any three from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1. See diagram given below and answer the following questions:



- | | | |
|----|---|---|
| a) | Define 1 weber. | 1 |
| b) | Indicate the magnetic field direction of the points R and S region which is produced by the PQ wire. | 2 |
| c) | Determine the amount of force acting on the PQ wire. | 2 |
| 2. | a) Prove that "The magnetic field is zero at the point inside the conducting wire produced by itself". | 2 |
| | b) A straight horizontal segment of copper wire carries $i = 28\text{A}$. What are the magnitude and direction of the magnetic field needed to float the wire, that is, to balance its weight? Its linear mass density is 46.6g/m . | 3 |
| 3. | a) Define 1 Coulomb. | 1 |
| | b) Write down the limitation of the Coulomb's law. | 2 |
| | c) Calculate the repulsive Coulomb force exist between two protons in a nucleus of iron. Assume a separation of $4 \times 10^{-15}\text{m}$. | 2 |
| 4. | a) State Gauss law. | 2 |
| | b) Formulate the Coulomb's law from Gauss law. | 3 |
| 5. | a) Write a short note "Equipotential Surface". | 2 |
| | b) Three charges $q_1 = -5 \times 10^{-8}\text{coul}$, $q_2 = 2 \times 10^{-8}\text{coul}$, $q_3 = 3 \times 10^{-8}\text{coul}$ are placed respectively at the three corners of an equilateral triangle; the length of any side 'a' of it is 10 cm. Calculate the potential energy of the system. | 3 |
| 6. | a) What is resistivity? | 1 |
| | b) Draw the electric lines of force for two opposite charges placed in nearer. | 2 |
| | c) What is the magnitude of the electric field strength E such that an electron, placed in the field would experience an electrical force equal to its weight? (Mass of the electron= $9.1 \times 10^{-31}\text{kg}$) | 2 |